

KARASUK, Russian Federation

WMO: 298140

Lat: 53.7345N

Lon: 78.0137E

Elev: 115

StdP: 99.95

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -34.7	(c) -31.6	(d) -37.8	(e) 0.1	(f) -34.7	(g) -34.8	(h) 0.1	(i) -31.5	(j) 12.1	(k) -7.6	(l) 10.4	(m) -6.7	(n) 1.9	(o) 240	(p) 0.610

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.9	31.0	18.4	29.4	18.1	27.7	17.5	20.4	27.0	19.5	25.9	18.8	25.1	3.9	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.3	13.4	22.8	17.3	12.6	22.2	16.4	11.9	21.7	59.5	27.0	56.4	25.8	53.9	25.0	23.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 10.3	(b) 8.8	(c) 7.7	(e) -37.1	(f) 34.3	(g) 4.0	(h) 1.6	(i) -40.0	(j) 35.5	(k) -42.3	(l) 36.4	(m) -44.5	(n) 37.3	(o) -47.4	(p) 38.5		
(a) 10.3	(b) 8.8	(c) 7.7	(e) -37.0	(f) 21.7	(g) 3.8	(h) 1.1	(i) -39.7	(j) 22.5	(k) -42.0	(l) 23.1	(m) -44.1	(n) 23.8	(o) -46.9	(p) 24.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	2.8	-18.0	-15.6	-7.1	5.3	13.4	19.0	20.4	18.0	12.1	4.3	-6.0	-13.9		
	DBStd	14.87	8.04	7.23	6.47	5.80	5.36	4.11	3.32	3.75	4.67	4.98	7.47	7.72		
	HDD10.0	3746	868	716	529	163	29	1	0	1	31	187	480	740		
	HDD18.3	5894	1127	949	788	392	168	42	18	53	194	436	730	998		
	CDD10.0	1102	0	0	0	22	134	271	323	248	93	10	0	0		
	CDD18.3	208	0	0	0	0	15	62	82	42	7	0	0	0		
	CDH23.3	1968	0	0	0	8	195	567	692	404	101	1	0	0		
	CDH26.7	570	0	0	0	1	49	173	204	114	29	0	0	0		
(14) Wind	WSAvg	3.4	3.5	3.6	3.8	3.8	3.7	3.2	2.8	2.8	3.1	3.4	3.7	3.9		
(15) Precipitation	PrecAvg	317	13	11	14	16	23	42	66	41	24	27	21	19		
	PrecMax	509	29	27	43	45	64	128	155	104	52	64	46	40		
	PrecMin	196	1	0	2	1	1	5	25	7	5	3	12	8		
	PrecStd	71	7	7	9	11	14	28	37	21	13	17	10	8		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.1	0.9	7.7	24.3	31.0	32.7	33.0	32.5	30.4	20.8	9.4	1.7		
		MCWB	0.4	-0.2	4.1	13.8	16.4	18.5	19.6	18.9	16.7	11.4	6.3	0.4		
	2%	DB	-1.5	-0.6	4.2	20.7	27.8	30.6	30.6	29.7	26.1	17.4	6.7	0.4		
		MCWB	-2.3	-1.5	2.0	12.0	15.0	17.9	19.3	18.2	15.0	10.1	4.8	-0.3		
	5%	DB	-3.9	-2.6	2.6	17.5	25.6	28.7	29.0	27.3	23.0	14.2	4.3	-1.3		
		MCWB	-4.5	-3.4	1.0	10.1	14.3	17.5	19.1	17.4	13.5	8.6	2.7	-2.1		
	10%	DB	-5.9	-4.8	1.2	14.6	22.9	26.7	27.2	25.4	20.0	11.5	2.4	-3.1		
		MCWB	-6.5	-5.5	0.0	8.5	13.3	17.2	18.4	16.6	12.5	7.3	1.1	-3.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.3	0.0	4.3	14.4	17.7	21.0	22.0	21.0	17.9	12.3	7.0	0.7		
		MCDB	0.9	0.7	6.9	23.4	27.9	28.7	28.6	27.4	27.6	18.2	8.3	1.3		
	2%	WB	-2.2	-1.5	2.4	12.5	16.2	19.6	20.8	19.4	15.7	10.8	4.7	-0.2		
		MCDB	-1.5	-0.6	4.0	19.5	25.4	26.6	27.5	26.2	23.9	16.1	6.6	0.4		
	5%	WB	-4.5	-3.4	1.1	10.6	15.0	18.8	20.0	18.4	14.4	9.2	2.8	-2.1		
		MCDB	-3.9	-2.6	2.4	16.4	23.3	25.6	26.1	25.4	21.6	13.5	4.2	-1.4		
	10%	WB	-6.5	-5.5	0.1	8.9	13.9	18.0	19.3	17.5	13.1	7.6	1.2	-3.9		
		MCDB	-5.9	-4.8	1.2	13.9	22.0	24.7	25.1	23.9	19.2	11.2	2.3	-3.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.9	8.5	8.3	9.8	11.3	10.3	9.9	10.7	10.5	8.3	6.5	7.6		
		MCDBR	8.3	7.9	7.3	14.1	14.7	13.4	12.4	13.9	15.7	13.3	6.9	7.8		
		MCWBR	8.4	7.6	6.0	7.4	5.7	4.7	4.2	5.0	6.6	7.1	5.3	7.8		
		MCDBR	8.2	7.6	7.0	13.5	12.9	10.9	10.5	11.7	13.7	11.9	6.8	7.8		
	5% WB	MCWBR	8.3	7.4	6.0	7.7	5.9	4.6	4.2	4.9	6.4	7.0	5.5	7.8		
(40) Clear-Sky Solar Irradiance	taub		0.294	0.325	0.382	0.405	0.408	0.413	0.439	0.421	0.373	0.352	0.330	0.298		
	taud		2.029	1.979	1.926	2.159	2.208	2.278	2.207	2.226	2.310	2.253	2.101	1.991		
	Ebn at Noon		624	731	768	819	839	838	805	796	791	711	574	521		
	Edh at Noon		75	110	145	132	133	125	132	123	99	84	70	64		
(44) All-Sky Solar Radiation	RadAvg		1.09	2.19	3.71	4.71	5.81	6.21	5.61	4.77	3.41	1.78	0.86	0.72		
	RadStd		0.09	0.10	0.29	0.34	0.48	0.42	0.38	0.36	0.31	0.22	0.11	0.09		

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	DBAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (12 neighbors)	DBAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page